M378K: March 23rd, 2026.

Mean Squared Error.

Review: • $\text{bias}(\hat{\theta}) = \mathbb{E}_{\hat{\theta}}[\hat{\theta}] - \theta$

• $\hat{\theta}$ is unbiased for θ if

$$\boxed{\text{bias}(\hat{\theta}) = 0 \quad \text{for all } \theta}$$

Example. • $Y_1, \dots, Y_n$ random sample from a dist'n w/ mean μ.

$\bar{Y}$ is unbiased for μ

• $Y_1, \dots, Y_n$ random sample from $N(\mu, \sigma)$ w/ both parameters unknown

$$\boxed{S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (Y_i - \bar{Y})^2}$$ is unbiased for σ^2

Def'n. Let $\hat{\theta}$ be an estimator for the parameter θ:

1. the error of $\hat{\theta}$ is $\hat{\theta} - \theta$;
2. the absolute error of $\hat{\theta}$ is $|\hat{\theta} - \theta|$;
3. the squared error of $\hat{\theta}$ is $(\hat{\theta} - \theta)^2$;
4. the mean squared error of $\hat{\theta}$ is

$$\boxed{MSE(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2]}$$

Proposition.

$$\boxed{MSE(\hat{\theta}) = (\text{bias}(\hat{\theta}))^2 + Var[\hat{\theta}]}$$

$\longrightarrow$:

$$MSE(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2]$$
$$= \mathbb{E}\left[\left((\hat{\theta} - \mathbb{E}[\hat{\theta}]) + (\mathbb{E}[\hat{\theta}] - \theta)\right)^2\right]$$

$$\mathrm{MSE}(\hat{\theta}) = \boxed{\mathbb{E}\left[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2\right]} + \cancel{\boxed{2\,\mathbb{E}\left[(\hat{\theta} - \mathbb{E}[\hat{\theta}])(\mathbb{E}[\hat{\theta}] - \theta)\right]}} + \boxed{\mathbb{E}\left[(\mathbb{E}[\hat{\theta}] - \theta)^2\right]}$$

$$= \mathrm{Var}[\hat{\theta}] + (\mathrm{bias}(\hat{\theta}))^2$$

deterministic

<u>Focus on</u>: $\mathbb{E}\left[(\hat{\theta} - \mathbb{E}[\hat{\theta}])(\mathbb{E}[\hat{\theta}] - \theta)\right] =$

$$= (\mathbb{E}[\hat{\theta}] - \theta)\,\underbrace{\mathbb{E}[\hat{\theta} - \mathbb{E}[\hat{\theta}]]}_{\underbrace{\mathbb{E}[\hat{\theta}] - \mathbb{E}[\hat{\theta}]}_{=0}}$$

$\square$

<u>Def'n</u>. The standard error of $\hat{\theta}$ is

$$\boxed{\mathrm{SE}(\hat{\theta}) = \sqrt{\mathrm{Var}[\hat{\theta}]}}$$

M378K Introduction to Mathematical Statistics
Problem Set #15
Bias. MSE.

Problem 15.1. Source: "Mathematical Statistics with Applications" by Wackerly, Mendenhall, Scheaffer.
Let $Y_1, , Y_2, Y_3$ be a random sample from $E(\tau)$. Consider the following five estimators of τ:

$$\hat{\theta}_1 = Y_1, \quad \hat{\theta}_2 = \frac{Y_1 + Y_2}{2}, \quad \hat{\theta}_3 = \frac{Y_1 + 2Y_2}{3}, \quad \hat{\theta}_4 = 3Y_{(1)}, \quad \hat{\theta}_5 = \bar{Y}.$$

Which ones of these estimators are unbiased? Among the unbiased ones which one has the smallest variance?

$\longrightarrow$: $\mathbb{E}[\hat{\theta}_1] = \mathbb{E}[Y_1] = \tau$ ✓

$\mathbb{E}[\hat{\theta}_2] = \mathbb{E}\left[\frac{Y_1+Y_2}{2}\right] = \tau$ ✓

$\mathbb{E}[\hat{\theta}_3] = \mathbb{E}\left[\frac{Y_1+2Y_2}{3}\right] = \tau$ ✓

$\mathbb{E}[\hat{\theta}_4] = \mathbb{E}[3 \cdot Y_{(1)}] = 3 \cdot \mathbb{E}[\underbrace{Y_{(1)}}_{\sim E(\tau/3)}] = 3 \cdot \frac{\tau}{3} = \tau$ ✓

$\mathbb{E}[\hat{\theta}_5] = \mathbb{E}[\bar{Y}] = \tau$ ✓

$Var[\hat{\theta}_1] = Var[Y_1] = \tau^2$

$Var[\hat{\theta}_2] = Var\left[\frac{Y_1+Y_2}{2}\right] = \frac{\tau^2}{2}$

$Var[\hat{\theta}_3] = Var\left[\frac{1}{3}(Y_1+2Y_2)\right] = \frac{1}{9}\left(\overbrace{Var[Y_1]}^{\tau^2} + 4 \cdot \overbrace{Var[Y_2]}^{\tau^2}\right) = \frac{5}{9}\tau^2$

$Var[\hat{\theta}_4] = Var[3 \cdot Y_{(1)}] = 9 \cdot Var[Y_{(1)}] = 9 \cdot \frac{\tau^2}{9} = \tau^2$

$Var[\hat{\theta}_5] = Var[\bar{Y}] = \boxed{\frac{\tau^2}{3}}$

Remark: When we want to estimate the mean,
$\mathbb{E}[\bar{Y}] = \text{mean}$, i.e., unbiased

$\Rightarrow MSE[\bar{Y}] = Var[\bar{Y}] = \frac{Var[Y_1]}{n} \Rightarrow SE[\bar{Y}] = \frac{SD[Y_1]}{\sqrt{n}}$ $\square$